BETHELMIND DIGITAL SOLUTIONS

CAC CERTIFIED ENTERPRISE (REG: 9912717) • CAC CERTIFIED ENTERPRISE (REG: 9912717) • LAGOS COMMERCIAL SUITE

OFFICIAL SOLAR BOQ LOAD CERTIFICATE

BM-SOLAR-BOQ-002015

VALIDATED ENGINEERING / COMMERCIAL OUTPUT

Solar & Renewable Energy BOQ Engineering Specification

Clean Energy & Inverter Engineering

VERIFIED SPECIFICATION

1. EXECUTIVE MATHEMATICAL & FINANCIAL BENCHMARKS

RECOMMENDED INVERTER CAPACITY
5.0 kVA / 48V Pure Sine Wave Hybrid

LITHIUM BATTERY STORAGE (LIFEPO4)
10.24 kWh (2x 5.12kWh 100Ah Rackmount)

SOLAR PV ARRAY REQUIRED
3.6 kWp (6x 600W Tier-1 Monocrystalline)

ESTIMATED TURNKEY INSTALLATION
₦4,850,000 NGN

DISCO BAND A GRID TARIFF HEDGE
₦185,000 / month

DIESEL GEN PAYBACK PERIOD
14.2 Months (vs ₦1,250/L PMS/AGO)

2. DETAILED COMMERCIAL BILL OF QUANTITIES / LINE ITEM SCHEDULE

SYSTEM COMPONENT	ENGINEERING SPECIFICATION	BRAND & WARRANTY LEVEL	ESTIMATED UNIT COST
Hybrid Solar Inverter	5kVA 48V Pure Sine Wave, Dual MPPT 120-450VDC	Growatt / Deye (5 Years Warranty)	₦1,450,000
Lithium LiFePO4 Battery	51.2V 200Ah (10.24kWh), 6000 Cycles @ 80% DoD	Felicity / Pylontech (5 Years Warranty)	₦2,100,000
Monocrystalline PV Panels	6x 600W N-Type Bifacial Tier-1 High-Efficiency	Jinko / Longi (25 Years Linear Yield)	₦780,000
Protection & Accessories	AC/DC SPD, 125A DC Breakers, 16mm Grounding	Schneider / Suntree (Statutory Surge Kit)	₦280,000
Certified Installation	Roof mount, trunking, load balancing & testing	COREN / Bethelmind Certified Engineers	₦240,000

3. NIGERIAN ROADSIDE & TECHNICAL FRAUD PREVENTION DIRECTIVES

AUDIT POINT	THREAT LEVEL	INSPECTION DIRECTIVE & FRAUD MITIGATION ACTION
Battery Chemistry Fraud Warning	HIGH THEFT RISK	Never accept rebranded lead-acid batteries disguised in lithium metal casings. Inspect CAN-bus BMS telemetry via RS485 communication.
Inverter Overloading Spike Check	EQUIPMENT FAILURE	Calculate starting inductive surge from 1.5HP inverter ACs and deep freezers. Ensure inverter handles 2x surge for at least 5 seconds.
DC Cable Undersizing Danger	FIRE HAZARD	Enforce minimum 25mm2 to 35mm2 pure copper DC battery cables. Aluminum-clad copper (CCA) cables melt under 100A continuous draw.

OFFICIAL STAMPED VERIFICATION & GUARANTEE

Calculated via Nigerian Trade & Market Indices • CAC Certified Enterprise (Reg: 9912717)
Certified Date: 26 October 2026 • Settlement: Moniepoint MFB 6805375225

BETHELMIND DIGITAL SOLUTIONS
Technical Strategy & Commercial Desk
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